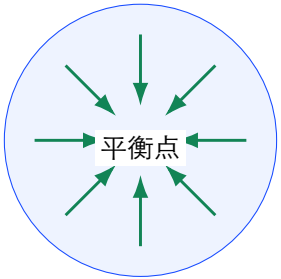
# 连续电荷区的各向同性平衡：体电荷允许三向曲率同号

$q\alpha > 0$ 时的恢复力




$\rho = -6\varepsilon_0\alpha$

$\nabla^2(\alpha r^2) = 6\alpha \Rightarrow \rho = -6\varepsilon_0\alpha$

$U = q\alpha r^2$；当 $q\alpha > 0$ 时原点三向稳定。

$\omega = \sqrt{2q\alpha/m}$；若 $q\alpha < 0$，原点三向均不稳定。